3.8.2026

Practical - 4

Setting up LAN

AIM:

To design, set up and configure local area network connecting 4 computers through network switch assign static IP address, verify connectivity using ping and demonstrate data sharing.

EQUIPMENT REQUIRED:

S.NO	Equipment	qty	Specifications
1.	Personal computer	4	Ethernet (NIC) enabled
2.	Network switch	1	8/16/24 port unmanaged/managed
3.	Ethernet cables	4	Straight - through RJ45 connectors.

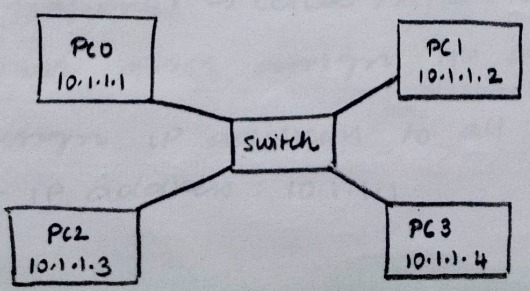
THEORY

LAN is a network that connects computer and devices within limited geographical area allowing them to share resources such as files, printers and internet.

Star topology is used here, where all PCs are individually connected to a central network switch. The switch operates at Layer 2 of OSI model and forwards data frames between connected devices based on their MAC addresses.

NETWORK DEVICES.

LAN was implemented using star topology



IP ADDRESS CONFIGURATION TABLE:

Device	IP Address	Subnet Mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0|1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0|2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0|3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0|4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

IMPLEMENTATION IN CISCO PACKET TRACER

Step 1 - Place devices

1. Open cisco Packet Tracer
2. From device category select End devices and drag 4 PCs
3. Select switches and drag 1 switch (2960 switch)

Step 2 - cable the Devices.

4. Click connections → select copper straight - through cable.
5. Click each PC, then click switch, select an available Fast Ethernet port
6. Repeat all 4 PC (ports Fa0\|1 to Fa0\|4). Green light confirms successful connection.

Step 3 - Assign IP Address to PCs

7. Click PC → Desktop tab → IP configuration.
8. Enter static IP address & subnet mask

Step 4 - Configure switch Management IP

9. Click switch → CLI
10. enable
    configure terminal
    interface vlan 1
    IP address 10.1.1.5 255.0.0.0
    no shutdown
    exit

Step 5 - Verify connectivity.

11. click PC → Desktop → command Prompt
12. use ping command to test connectivity to other PC
13. successful "Reply From..." message.

Step 6 - Demonstrate Data sharing

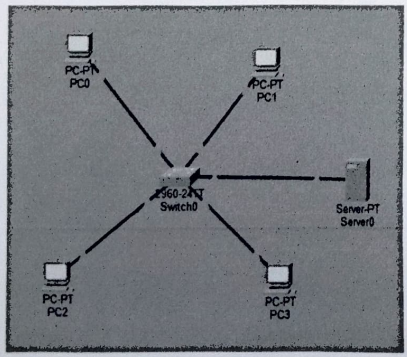
14. Add server device (End device → server PT) and connect to switch
15. Assign IP address via IP configuration
16. open config tab enable HTTP service
17. open web browser, enter server IP address to load page.

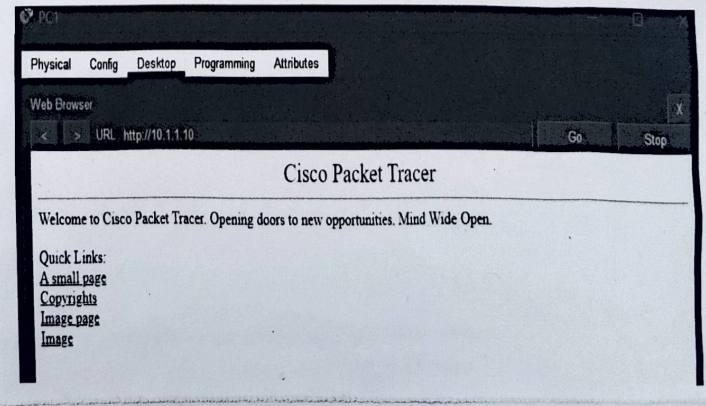
Step 7 - Observe Packet Flow

18. switch to simulation mode
19. send ping or web request between two devices and use Play/capture control to step through packet.

OBSERVATION:

All 4 PC successfully connected to the switch in star topology and assigned unique static IP address within same subnet (10.0.0.0 | 255.0.0.0). The shared folder successfully accessed confirming LAN was function correctly for resource sharing.





RESULT:

The lab was successfully designed and configured. connectivity between all PCs verified using ping command and resource sharing was successfully demonstrated.